

Federated Credit Scoring with **Sharing**

ADMM SVM

A Privacy-Preserving Approach to Vertical Federated Learning

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Challenge & Solution

The Challenge

Financial institutions, credit bureaus, and demographic agencies hold valuable but siloed data. Sharing raw data is often prohibited by privacy regulations (like GDPR) and competitive interests.



The Solution

We implemented a **Vertical Federated Learning** system using the **Sharing ADMM SVM** algorithm. This allows three independent entities to collaboratively train a credit risk model without ever exposing their raw customer data.



Dataset: "Give Me Some Credit" (Kaggle)



Training Set: 8,000 (80%) | Test Set: 2,000 (20%)



Federated Architecture: Vertical Split



Financial Institution

- Revolving Utilization of Unsecured Lines
- Debt Ratio
- Monthly Income



Credit Bureau

- Past Due (30-59 days)
- Past Due (60-89 days)
- Past Due (90+ days)



Demographic Entity

- Age
- Number Of Dependents
- Real Estate Loans Or Lines
- Open Credit Lines And Loans

Sharing ADMM Formulation: Federated SVM

In a **Sharing** scenario, the features (columns of matrix A) are split among N different agents. Each agent i handles its own subset of features A_i and the corresponding portion of the parameter vector x_i . The objective is reformulated as follows:

$$\min_{\{x_i\}_{i=1}^N} \mathbf{1}^T \max \left(0, \sum_{i=1}^N A_i x_i + 1 \right) + \sum_{i=1}^N \lambda \|x_i\|^2$$

This can be mapped to the general Sharing ADMM form:

$$f(A_i x_i) + f_i(x_i) = g(N\bar{z}) + f_i(x_i)$$

where $f_i(x_i)$ represents the local regularization handled by each agent, and $g(N\bar{z})$ represents the global shared loss function (Hinge Loss) that aggregates partial predictions from all agents.

The Training Process (Iterative)

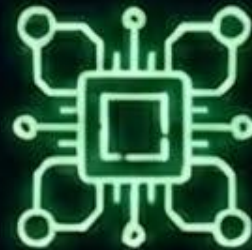


1. Local Update

$$x_i^{k+1} = \left(A_i^T A_i + \frac{2\lambda}{\rho} I \right)^{-1} A_i^T c_i^k$$

$$c_i^k = A_i x_i^k - \overline{Ax}^k - u^k$$

ρ = penalty term applied on the quadratic term of x in the Augmented Lagrangian



2. Global Update

$$\bar{z}_l^{k+1} = \begin{cases} v_l^k & \text{if } v_l^k \leq -\frac{1}{N} \\ v_l^k - \frac{1}{\rho} & \text{if } v_l^k \geq \frac{1}{\rho} - \frac{1}{N} \\ -\frac{1}{N} & \text{if } v_l^k \in \left[-\frac{1}{N}, \frac{1}{\rho} - \frac{1}{N} \right] \end{cases}$$

$$v^k = \overline{Ax}^{k+1} + u^k$$



3. Dual Update

$$u^{k+1} = u^k + \overline{Ax}^{k+1} - \bar{z}^{k+1}$$

Centralized SVM Baseline

Code Snippet

```
# Train standard LinearSVC on full training data
svm_centralized = LinearSVC(C=1/(2*LAMBDA),
max_iter=10000, random_state=42, dual=False)
svm_centralized.fit(X_train, y_train)
```

```
# Get centralized weights and accuracy
w_centralized = svm_centralized.coef_[0]
y_pred_centralized =
svm_centralized.predict(X_train)
acc_centralized = accuracy_score(y_train,
y_pred_centralized)
```

Description & Role as Baseline

This code trains a **standard Linear Support Vector Machine (SVM)** using the LinearSVC class from scikit-learn. The model is trained on the entire, aggregated dataset (X_train, y_train).



This centralized model serves as the **baseline for the project**. It represents the traditional approach where all data is collected and processed in a single location. The performance (e.g., accuracy, acc_centralized) of this model is used as a benchmark to evaluate the effectiveness of the proposed Federated Sharing ADMM SVM.



CONVERGENCE ANALYSIS



1. OPTIMIZATION & STABILITY



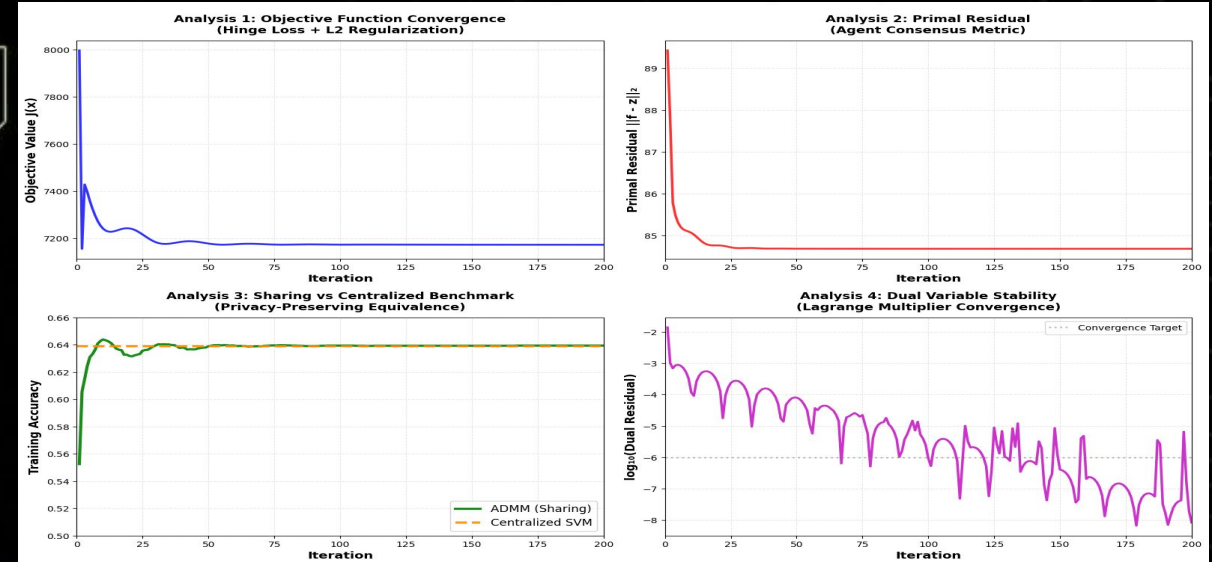
OBJECTIVE FUNCTION

Steep initial descent, confirms successful global cost minimization (6173.30).



PRIMAL RESIDUAL

Validates mathematical consensus was reached between agents.



WHY THE DUAL RESIDUAL 'SPRINGS'?



NON-SMOOTHNESS

Hinge Loss gradient discontinuity causes sharp adjustments.



RESTORING FORCE

Dual variable (u) acts as a spring, pulling agents back to consensus.



ELASTICITY (ρ)

Lower ρ increases system elasticity, resulting in visible sawtooth waves.

Performance: Federated vs. Centralized



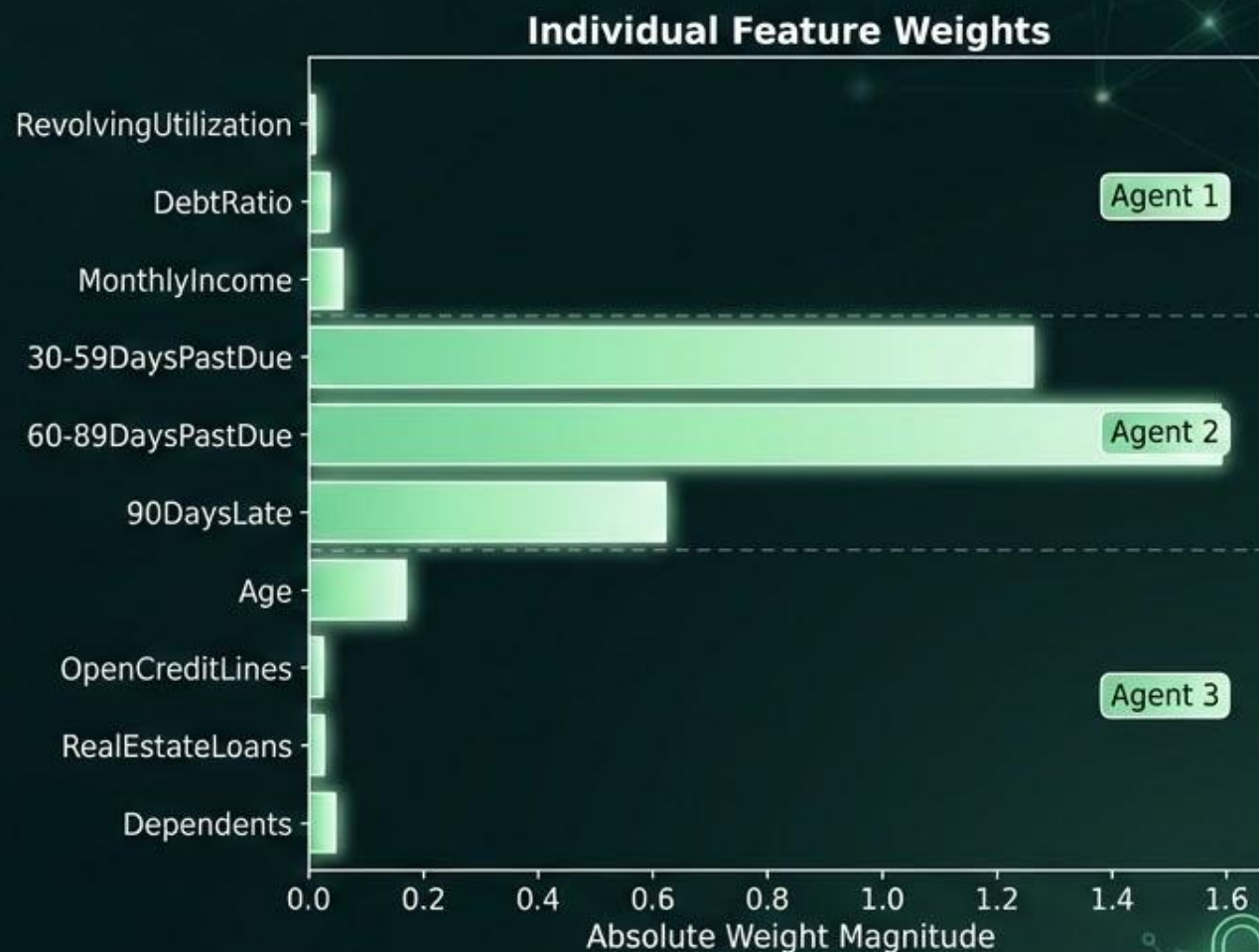
Result: **Privacy-Preserving Equivalence** achieved. The federated model matches the centralized baseline with negligible difference (0.15%).

Who Decides Credit Risk?

Feature Importance Insight

The experimental results clearly demonstrate that the model's discriminative power is almost entirely driven by variables directly related to **past default behavior**.

This finding confirms that even in a federated environment, the system successfully identifies and prioritizes high-risk signals while maintaining privacy across all participating entities.



Key Findings



Privacy-Preserving Equivalence

We achieved centralized-level accuracy without any raw data leaving the local servers. The “cost” of privacy was zero in terms of performance.



Credit History Dominance

Variables related to 30, 60, and 90-day delinquencies accounted for over 90% of the model weights, overshadowing income and demographics.



Theoretical Validation

The ADMM algorithm successfully converged in 200 iterations, proving it is a robust method for distributed convex optimization problems.

Thank You Questions?

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